

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 1

**YEAR 11 YEARLY EXAMINATION
SEPTEMBER 2008**

General Instructions

- Working time allowed – 90 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions
- All questions are of equal value

NAME : _____

TEACHER : _____

QUESTION 1	QUESTION 2	QUESTION 3	QUESTION 4	QUESTION 5	TOTAL

Question 1

(solutions must be in black or blue pen)

Marks

a) Sketch $y = (x + 1)^2(x - 3)$ clearly showing all intercepts.

2

b) Simplify $\frac{2 + \frac{1}{x}}{2x^2 + x}$

2

c) Solve $\frac{3x}{x+1} \leq 1$

2

d) Expand and simplify $(\cos A - \sin A)(\cos A + \sin A)$

2

expressing your answer in terms of $2A$.e) If α , β and δ are the roots of $2y^3 - 8y^2 + 3y + 1 = 0$

4

find the value of i) $\alpha + \beta + \delta$ ii) $\alpha\beta + \alpha\delta + \beta\delta$ iii) $\alpha\beta\delta$ iv) $\alpha^2 + \beta^2 + \delta^2$

Question 2 (Start a new page)

Marks

- a) Find the coordinates of the point that divides the interval from $(-4, 2)$ to $(6, 9)$ internally in the ratio $4:1$. 2
- b) Find the remainder when $x^4 - 2x^2 + 5$ is divided by $x^2 + x - 2$. 2
- c) Find the equation of the parabola with focus $(0, 6)$ and directrix $y = 2$. 2
- d) Find the locus of the set of points $P(x, y)$ which are equidistant from the points $A(1, 8)$ and $B(5, -2)$. 2
- e) i) Express $3\sin\theta - 2\cos\theta$ in terms of t where $t = \tan\frac{\theta}{2}$. 2
- ii) Hence, or otherwise solve $3\sin\theta - 2\cos\theta = 2$ for $0^\circ < \theta < 180^\circ$ (nearest degree) 2

Question 3 (Start a new page)

Marks

- a) A curve has parametric equations $x = \frac{t}{3}$, $y = 4t^2$. 1

Find the Cartesian equation for this curve.

- b) Find the acute angle between the lines $2x - 3y + 4 = 0$ 2
and $x + y - 6 = 0$ giving your answer correct to the nearest degree.

- c) Factorise fully $x^3 + 7x^2 + 2x - 40$ 3

- d) i) Sketch the region in the number plane defined by 2

$$y \leq |2x - 3|$$

- ii) For what values of m does the equation 1

$$|2x - 3| = mx \quad \text{have 2 solutions.}$$

- e) Find the value of a given that 1

$$x^3 - 2x^2 + a \equiv (x + 3)Q(x) + 3 \quad \text{where } Q(x) \text{ is a polynomial.}$$

- f) By rationalising the numerator, find the exact value of the following limit 2

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+5} - \sqrt{5}}{x}$$

Question 4 (Start a new page)

Marks

- a) i) Derive the equation of the tangent to the parabola $x^2 = 8y$ at the point $P(4p, 2p^2)$. 2
- ii) Find the point of intersection of the two tangents to the parabola $x^2 = 8y$ drawn from the points $P(4p, 2p^2)$ and $Q(4q, 2q^2)$. 2
- b) Write down the equation of a monic polynomial of degree 3, which is an odd function and has a root at $x = 3$. 2
- c) Find the value of m given that the equation $x^3 + mx + 2 = 0$ has a double root. 2
- d) If $f(x-1) = x^2 - 4x$ find $f(a)$ in simplest terms. 1
- e) i) Sketch $y = f(x)$ given 2
- $$f(x) = \begin{cases} x^2 + 1, & \text{for } x < 0 \\ 1 - x^2, & \text{for } 0 \leq x \leq 1 \\ x - 1, & \text{for } x > 1 \end{cases}$$
- ii) At which point or points is the function in part i) not differentiable. 1

Question 5 (Start a new page)

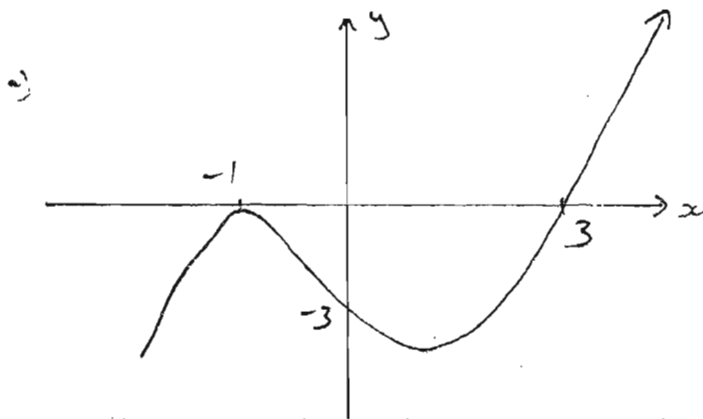
Marks

- a) Solve $3^{2x} - (1 + \sqrt{3})3^x + \sqrt{3} = 0$ 2
- b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.
- i) Find an expression in terms of a and p for the distance PS . 1
where S is the focus of the parabola.
- ii) Find the equation of the chord PQ . 2
- iii) If this chord passes through the point $M(0, 4a)$ show that $pq = -4$ 1
- iv) The chord PQ which passes through the point M meets 3
the directrix of $x^2 = 4ay$ at the point A .
Find the coordinates of A in terms of a and p .
- c) From a point A which lies due North of a tower and at ground level, 3
the top of the tower has an angle of elevation of 24° .
From a point B which lies due East of the same tower and is also at
ground level, the top of the tower has an angle of elevation of 34° .
Calculate the bearing of point B from point A .

End of Paper

SOLUTIONS

QUESTION 1



$$b) \frac{\frac{2x+1}{x}}{x(2x+1)} = \frac{1}{x^2}$$

$$c) \frac{3x}{x+1} \leq 1$$

$$-1 < x \leq \frac{1}{2}$$

$$d) \cos^2 A - \sin^2 A = \cos 2A$$

$$i) 4$$

$$ii) \frac{3}{2}$$

$$iii) -\frac{1}{2}$$

$$iv) (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = 4^2 - 2\left(\frac{3}{2}\right)$$

$$= 13$$

QUESTION 2

$$a) \left(4, 7\frac{3}{5}\right)$$

$$b) \begin{array}{r} x^2 - x + 1 \\ x^2 + x - 2 \overline{) x^4 - 2x^2 + 5} \\ \underline{x^4 + x^3 - 2x^2} \\ -x^3 \\ \underline{-x^3 - x^2 + 2x} \\ x^2 - 2x + 5 \\ \underline{x^2 + x - 2} \\ -3x + 7 \end{array}$$

$$\therefore \text{Remainder} = -3x + 7$$

$$c) x^2 = 8(y-4)$$

$$d) PA = PB$$

$$\sqrt{(x-1)^2 + (y-8)^2} = \sqrt{(x-5)^2 + (y+2)^2}$$

$$\begin{aligned} x^2 - 2x + 1 + y^2 - 16y + 64 &= x^2 - 10x + 25 + y^2 + 4y + 4 \\ 8x - 20y + 36 &= 0 \\ 2x - 5y + 9 &= 0 \end{aligned}$$

$$e) i) 3\left(\frac{2t}{1+t^2}\right) - 2\left(\frac{1-t^2}{1+t^2}\right)$$

$$= \frac{6t - 2 + 2t^2}{1+t^2}$$

$$ii) \frac{2t^2 + 6t - 2}{1+t^2} = 2$$

$$6t = 4$$

$$t = \frac{2}{3}$$

$$\therefore \theta = 67^\circ$$

QUESTION 3

a) $x = \frac{t}{3} \Rightarrow t = 3x$

$$y = 4t^2$$

$$\therefore y = 4(3x)^2$$

$$y = 36x^2$$

b) $m_1 = \frac{2}{3} \quad m_2 = -1$

$$\tan \theta = \frac{\left| \frac{2}{3} - (-1) \right|}{1 + \left(\frac{2}{3} \times -1 \right)}$$

$$= 5$$

$$\therefore \theta = 79^\circ$$

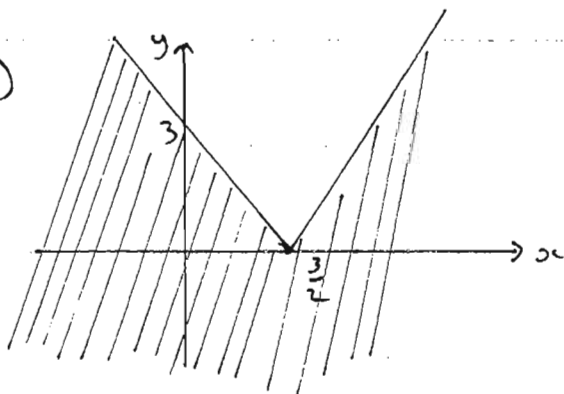
c) $P(2) = 8 + 28 + 4 - 40$
 $= 0$

$\therefore (x-2)$ is a factor

$$\therefore (x-2)(x^2 + 9x + 20)$$

$$= (x-2)(x+4)(x+5)$$

d) i)



ii) $0 < m < 2$

e) $P(-3) = 3$

$$-27 - 18 + a = 3$$

$$a = 48$$

f) $\frac{\sqrt{x+5} - \sqrt{5}}{x} \times \frac{\sqrt{x+5} + \sqrt{5}}{\sqrt{x+5} + \sqrt{5}}$

$$= \frac{x+5-5}{x(\sqrt{x+5} + \sqrt{5})}$$

$$= \frac{1}{\sqrt{x+5} + \sqrt{5}}$$

$$\therefore \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+5} + \sqrt{5}}$$

$$= \frac{1}{2\sqrt{5}}$$

QUESTION 4

a) i) $y = \frac{x^2}{8}$

$$\frac{dy}{dx} = \frac{2x}{8}$$

$$= \frac{x}{4}$$

when $x = 4p$

$$\frac{dy}{dx} = \frac{4p}{4}$$

$$m_T = p$$

$\therefore$ equation of tangent

$$y - 2p^2 = p(x - 4p)$$

$$y = px - 2p^2$$

$$\begin{aligned} \text{ii)} \quad y &= px - 2p^2 \\ y &= qx - 2q^2 \end{aligned}$$

$$\therefore px - qx - 2p^2 + 2q^2 = 0$$

$$x(p - q) = 2(p - q)(p + q)$$

$$x = 2(p + q)$$

$$\begin{aligned} \therefore y &= 2p(p + q) - 2p^2 \\ &= 2pq \end{aligned}$$

$$\therefore \text{pt is } (2(p + q), 2pq)$$

$$\text{b)} \quad p(x) = x(x - 3)(x + 3)$$

$$\text{c)} \quad x^3 + mx + 2 = 0$$

let roots be α, α, β

$$\therefore 2\alpha + \beta = 0 \quad \text{①}$$

$$\alpha^2 + 2\alpha\beta = m \quad \text{②}$$

$$\alpha^2\beta = -2 \quad \text{③}$$

Solving ① and ③ simultaneously

$$\beta = -2\alpha \Rightarrow \alpha^2(-2\alpha) = -2$$

$$\alpha^3 = 1$$

$$\alpha = 1$$

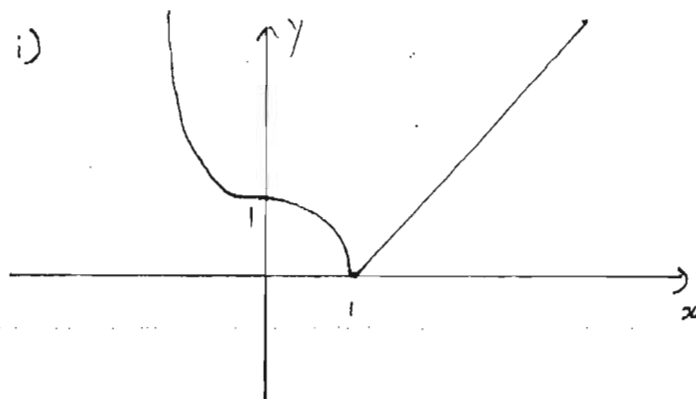
$$\therefore \beta = -2$$

$$\therefore m = -3$$

$$\text{d)} \quad f(a) = (a+1)^2 - 4(a+1)$$

$$= a^2 - 2a - 3$$

e) i)



ii) not differentiable at $x = 1$.

QUESTION 5

$$\text{a)} \quad \text{let } u = 3^x$$

$$u^2 - (1 + \sqrt{3})u + \sqrt{3} = 0$$

$$(u - 1)(u - \sqrt{3}) = 0$$

$$u = 1, \sqrt{3}$$

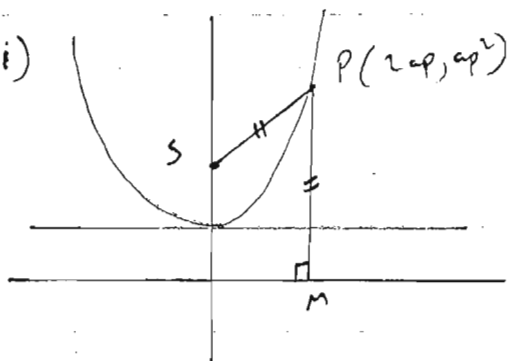
$$\therefore 3^x = 1$$

$$x = 0$$

$$3^x = \sqrt{3}$$

$$x = \frac{1}{2}$$

b) i)



$$\begin{aligned} PS &= PM \quad (\text{by definition}) \\ &= \sqrt{op^2 + a} \end{aligned}$$

$$i) m_{pa} = \frac{qp^2 - aq^2}{2ap - 2aq}$$

$$= \frac{p+q}{2}$$

equation

$$y - ap^2 = \left(\frac{p+q}{2} \right) (x - 2ap)$$

$$y = \frac{1}{2}(p+q)x - apq$$

$$iii) \text{ sub } (0, 4a)$$

$$4a = \frac{1}{2}(p+q)0 - apq$$

$$4a = -apq$$

$$pq = -4$$

$$iv) \text{ sub } y = -a \text{ into chord}$$

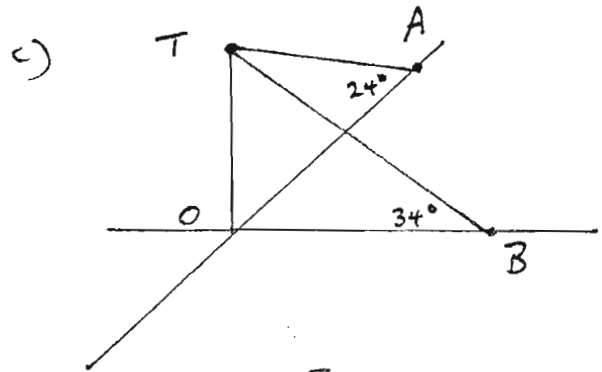
$$-a = \frac{1}{2}(p+q)x - apq$$

$$x = \frac{2a(pq-1)}{p+q}$$

$$\text{but } q = -\frac{4}{p}$$

$$= \frac{2a(-5)}{p - \frac{4}{p}}$$

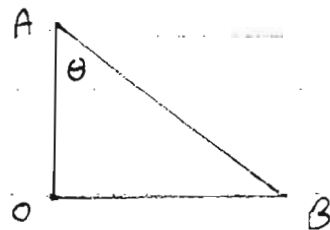
$$= \frac{-10ap}{p^2 - 4}$$



$$\tan 24^\circ = \frac{OT}{OA}$$

$$\therefore OA = \frac{OT}{\tan 24^\circ}$$

$$\text{also } OB = \frac{OT}{\tan 34^\circ}$$



$$\tan \theta = \frac{OB}{OA}$$

$$= \frac{OT}{\tan 34^\circ} \div \frac{OT}{\tan 24^\circ}$$

$$= \frac{\tan 24^\circ}{\tan 34^\circ}$$

$$\theta = 33^\circ 26'$$

$$\therefore \text{bearing is } 146^\circ 34'$$